

RENEWABLE ENERGY

PRESENTED BY ZAVIER LEE

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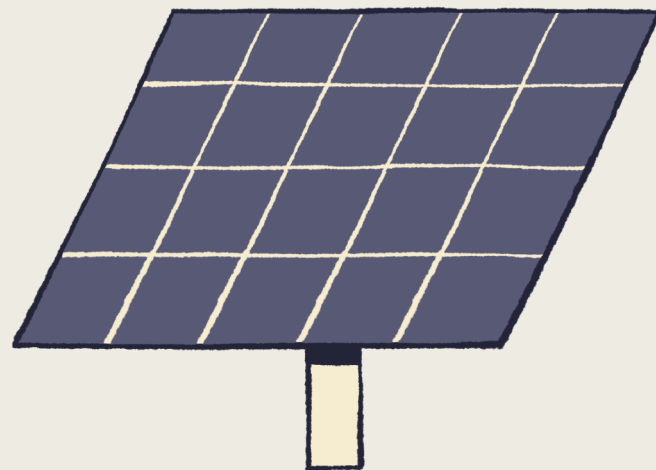
16-17 TARGETED COMPANIES

THE ECOSYSTEM

(03)

THE SUPPLIER

TECHNOLOGY OEMS & MODULE SUPPLIERS



- China holds more than 80% share
- Trends of commoditization and oversupply

- JinkoSolar (China)
- Longi Green Technology (China)
- First Solar (US)

THE BUILDER

ENGINEERING, PROCUREMENT, CONSTRUCTION AND COMMISSIONING (EPCC)



- Capture value from contract margin, material cost advantages and project delivery efficiency.
- Capital intensive and running on tight cash flow.

- Solarvest (Malaysia)
- Cypark Resources (Malaysia)
- Sunview (Malaysia)

THE OPERATOR &/ OWNER

INDEPENDENT POWER PRODUCER (IPP)



- Own and operate RE assets, selling electricity to utilities and large users.
- They manage operational risks.

- Malakoff
- YTL Power
- Edra Energy

THE ECOSYSTEM

(04)

THE EQUITY OWNER

THE STABILISER

THE BUYER

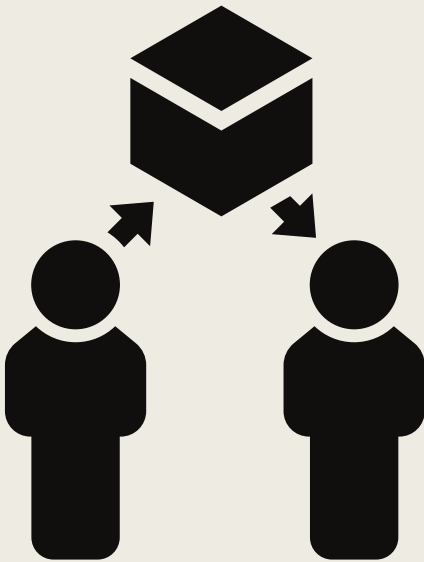
ASSET OWNER



GRID-SCALE ENERGY STORAGE (BESS) DEVELOPERS/OPERATORS



OFFTAKER



- Generate returns by securing predictable cash flows and capturing dividend distributions proportional to equity ownership

- Capture value from margin spread, maintenance of grid stability, capacity commitment.
- Value is also created from cost savings such as peak shaving and compliant cost

- Enter into long term power purchase agreements to lock in tariff rate.
- Buyers from Utilities / Corporate Consumers / Neighboring Countris

- Gentari (Subsidiary of Petronas)
- Tenaga Nasional Berhad
- Funds Operator

- Solarvest (Malaysia)
- CATL (China)
- Gotion (China)

- Tenaga Nasional Berhad
- Hyperscale Data Center
- Singapore

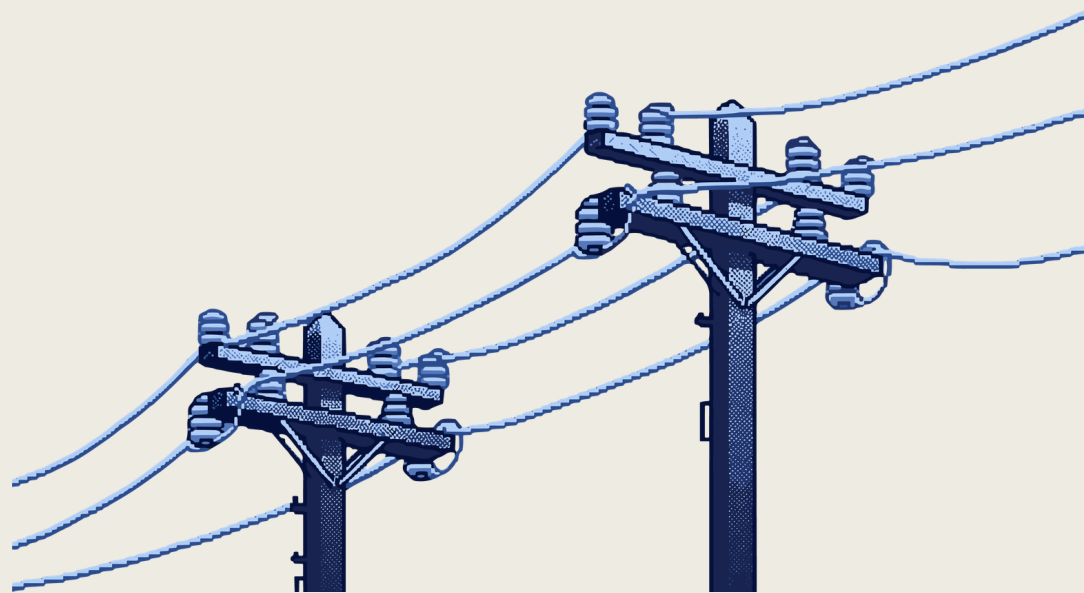
THE ECOSYSTEM

(05)

THE DISTRIBUTOR

THE STABILISER

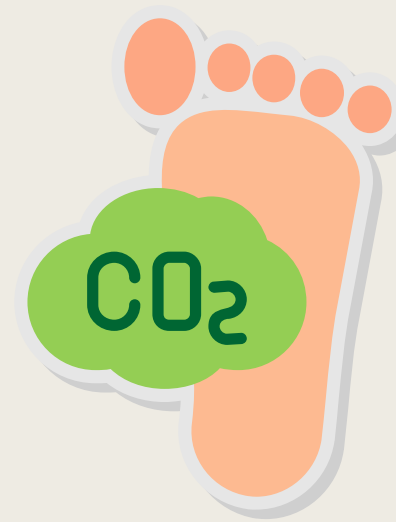
GRID OPERATOR



- Monopoly players in Malaysia.

- Tenaga Nasional Berhad
- Sarawak Energy Berhad
- Sabah Electricity Sdn Bhd

CARBON MARKET & CERTIFICATION BODIES



- ESG monetization and selling of credit to companies wishing to offset their carbon footprint.
- Stringent regulatory mandates have catalyzed growth in this sector

- Bursa Carbon Exchange
- Cenergi SEA
- Saxon Renewables (Sub. of Solarvest)

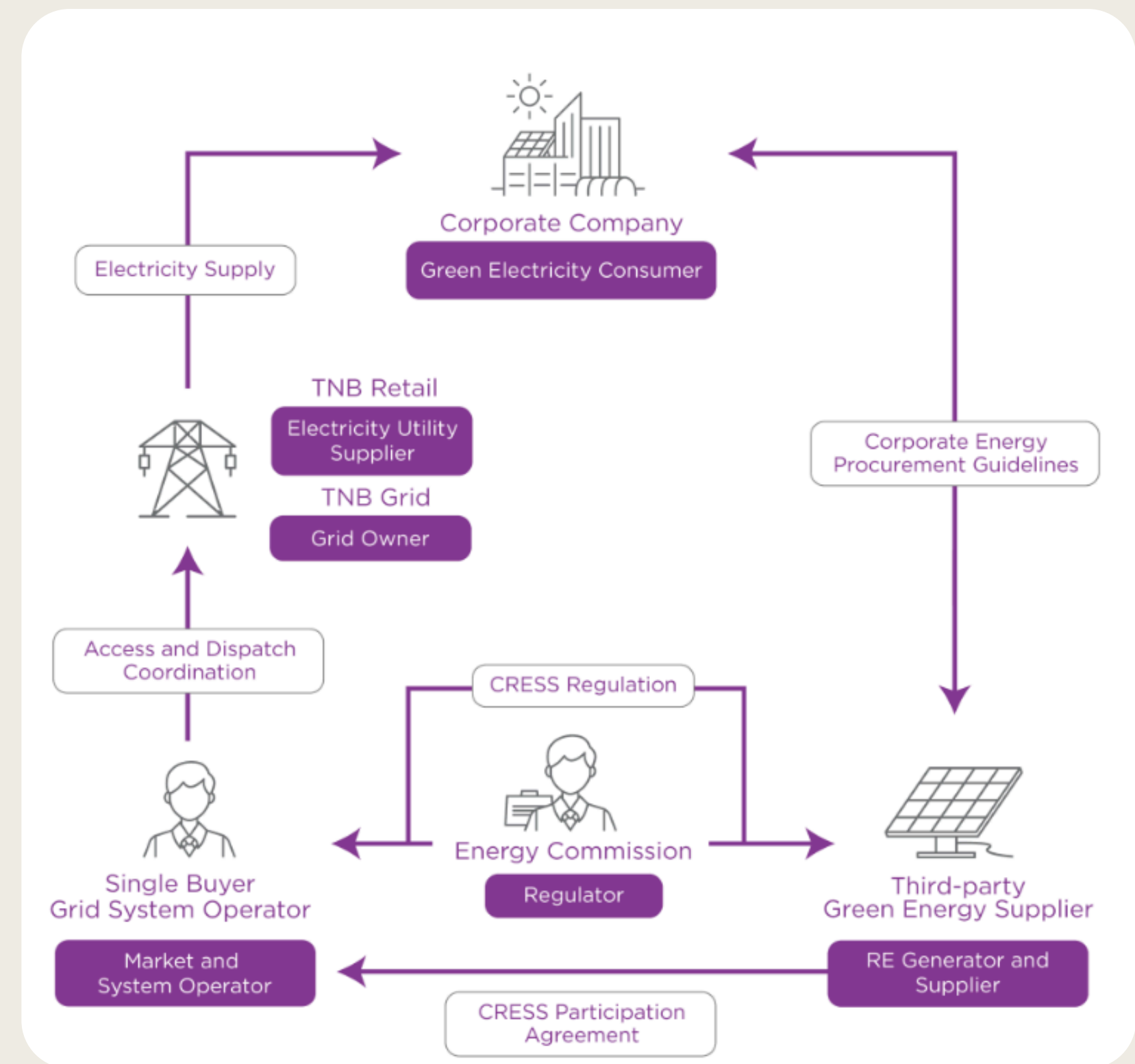
LOCAL DRIVER

Corporate Renewable Energy Supply Scheme (CRESS)

Model	Open Grid Access System where TNB acts as the toll road
Buyer	Corporate Consumers
Seller	Third Party Renewable Energy Producer

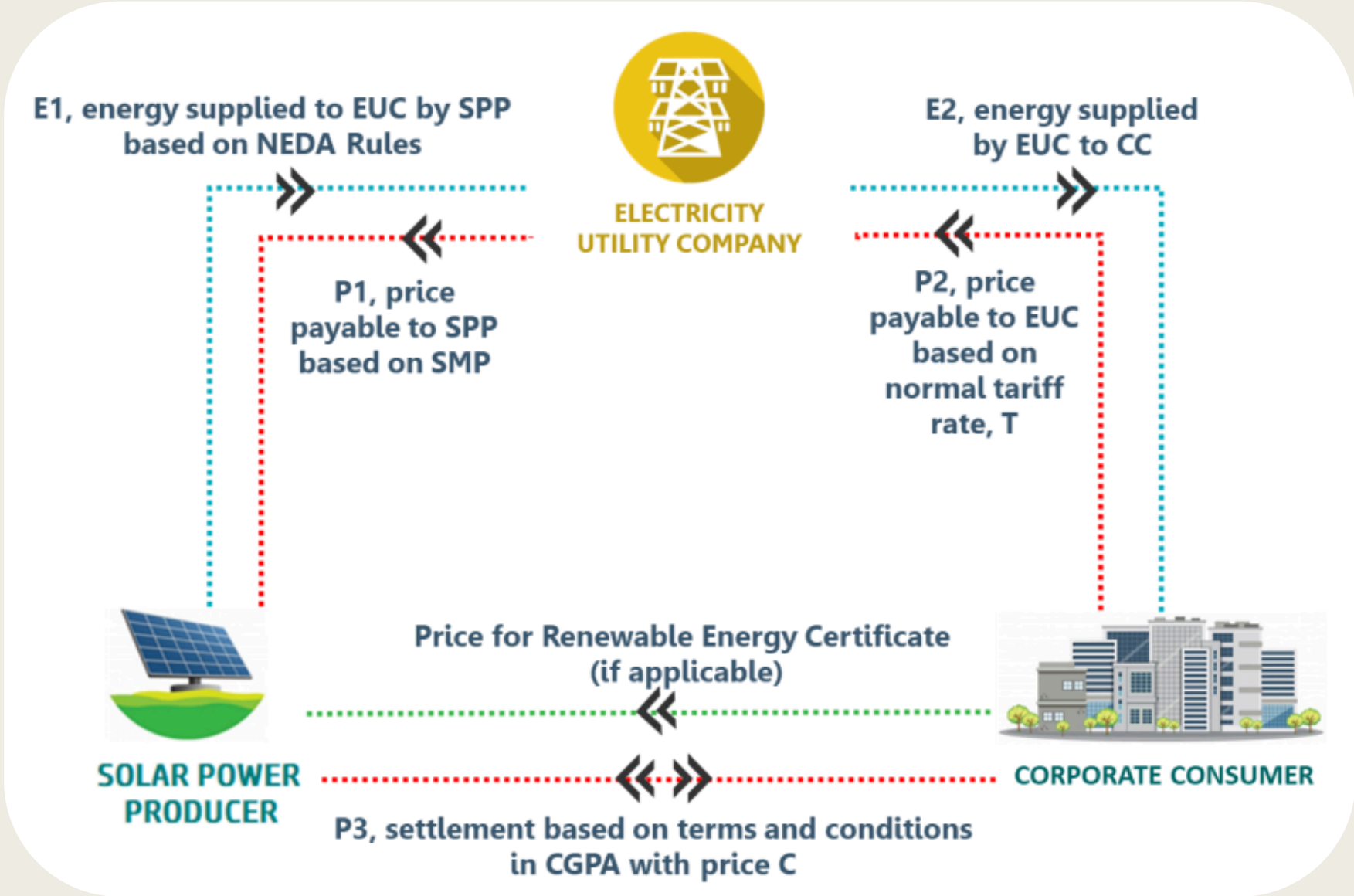
Market Activity:

DayOne Data Centers:	TNB Renewable to supply up to 500 MW of solar backed RE for over 21 years
Gentari & Gamuda	supply ~1.5 GW of solar power with battery storage primarily for hyperscale data centers.



Source: Solarvest

LOCAL DRIVER



Source: Energy Commission

** If the tariff rate paid by the corporate consumers is higher than the CGPA price, the consumer will pay for the positive difference to the solar developer and vice versa.*

Corporate Green Power Programme (CGPP)

Model	Virtual PPA without physical delivery of energy
Buyer	Corporate Consumers
Seller	Third Party Renewable Energy Producer

Market Activity:

Solarvest	secured 5 CGPA with 59.98 MWac electricity demand for 21 years
TotalEnergies & MK Land	To supply Google with 1 TWh of renewable energy to its data centers in Malaysia

LOCAL DRIVER

(08)

Large Scale Solar (LSS)

Model	Competitive bidding to win contract
Buyer	TNB
Seller	Third Party Renewable Energy Producer

LSS 5/LSS Petra 5+ Winning Bidders

Companies (JV)	Capacity	Control	Est. Value
Malakoff & Solarvest	470 MWac	80%/20%	\$425 million
Sunview & Cypark	99.99 MWac	51%/49%	91 million
.....			
<u>EPCC Contract</u>			
Founder Group	9.5 MW	-	\$8.6 million

LSS Cycle	Year Launched	Total Quota (MWac)	Estimated Total Investment (CAPEX)	Theme / Name
LSS 1	2016	250 MW	~RM 1.5 Billion	Operational
LSS 2	2017	460 MW	~RM 2.2 Billion	Operational
LSS 3	2019	500 MW	~RM 2.0 Billion	Operational
LSS 4	2020	1,000 MW	~RM 4.0 Billion	Operational
LSS 5	2024	2,000 MW	~RM 8.0 Billion	Developing
LSS Petra 5+	2025	1975 MW	~RM 6.0 billion	Pending

EXTERNAL DRIVER

ASEAN Power Grid (APG)

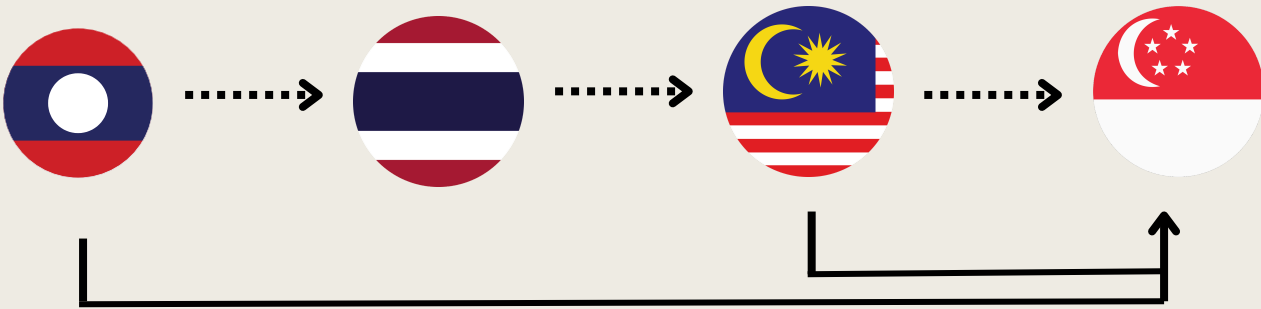
7.7 GW ➤ > 15.4 GW



\$764 billion of investment need

Nearly double the size of Malaysian GDP

LTMS-PIP (Lao PDR–Thailand–Malaysia–Singapore Power Integration Project)



2022-2024: 100 MW
2024-2026: 200 MW

BIMP-PIP (Brunei-Indonesia-Malaysia-Philippines Power Integration Project)

Under feasibility studies

Vietnam-Malaysia-Singapore Tripartite Alliance (May 2025)

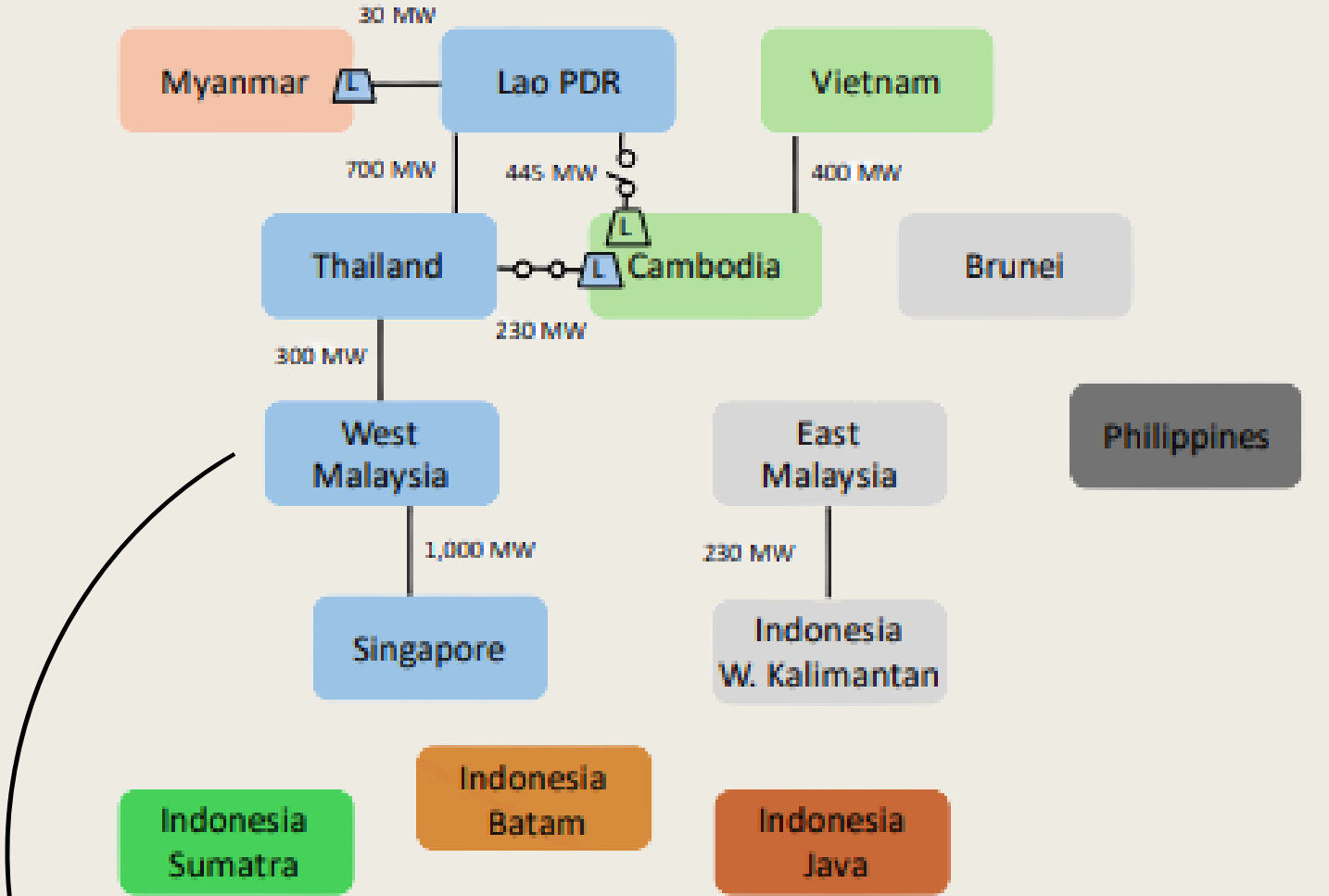
TNB and PETRONAS (Malaysia), PetroVietnam Technical Services Corporation (Vietnam), and Sembcorp (Singapore) to develop a 2,000-megawatt offshore wind energy project

EXTERNAL DRIVER

ASEAN Power Grid (APG)

Generator	- Laos - Vietnam - Indonesia
Transit/Growing generator	- Malaysia - Thailand
Offtaker	- Singapore - Brunei (Emerging)

(10)



Sit closest to the largest RE buyer in ASEAN



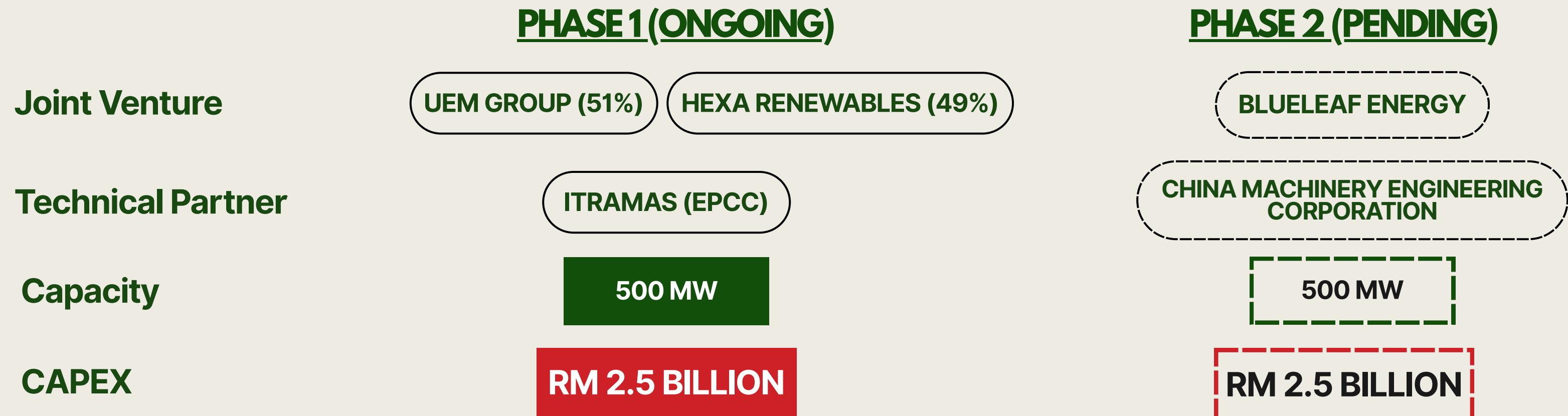
Greater grid capacity
Reliance on energy storage system (BESS)

2023

RM 7 billion sukuk to finance UEM's green initiatives

2024

40-Acre RE Industrial Park - **1GW** hybrid solar photovoltaic (PV)



PAST DEALS UEM GROUP BHD

PAST DEALS GENTARI (SUB OF PETRONAS)

2023 **2,000MW** of solar PV installed capacity & **4,400MWh** of battery storage

Joint Venture



Countries Involved



CAPEX

\$ 3 BILLION

Supply

300MW OF NON-INTERMITTENT, ZERO-EMISSION ELECTRICITY

Technical Partners



PAST DEALS

BROOKFIELD ASSET MANAGEMENT

\$ 1 BILLION COMMITMENT VIA BROOKFIELD'S CATALYTIC TRANSITION FUND (CTF)

2023 Acquire Alba Renewables

Acquisition

BROOKFIELD ASSET MANAGEMENT (CANADA)

ALBA RENEWABLES (SINGAPORE)

Asset Involved

1.8 GW PORTFOLIO OF WIND, SOLAR, AND BATTERY STORAGE ASSETS

2023 Acquire **100 MW** operational wind project in Vietnam

2025 Form partnership with Solarvest

Joint Investment

BROOKFIELD ASSET MANAGEMENT (CANADA)

SOLARVEST (MALAYSIA)

Deliverable

1.5 GW OF UTILITY-SCALE SOLAR AND BATTERY ENERGY STORAGE PROJECTS

DEAL IDEA

(14)

Deal Type	Asset	Rationale	Seller	Buyer
Farm Down (Sell stakes in operating RE assets)	LSS 1 - 3 1.Danau Tok Uban LSS 2 (98 MWp) 2.Sik LSS 2 (49 MWp) 3.Merchang LSS 3 (172 MWp)	Developers (Seller) 1.Cypark has been facing project delays, declining revenue and increasing debt burden 2.Exacerbated by the heavy overhead cost of RE plant development where they have been bleeding cash for years 3.With their goal to expand their energy capacity, they need to first free up their financial capacity as their debt-to-asset ratio has grown from 0.4 (2018) to 0.74 (Current) 4.While they wish to build recurring income stream, hence, we do farm down to allow them to retain some equity stakes Funds (Buyers) 1.Looking for ESG compliant stable yield instruments 2.Long-term PPA secures 20+ years of stable income 3.Assets are already operational, hence without construction risk	Cypark Resources	KWAP - Dana Iklim (Target deployment of RM2 billion) Permodalan Nasional Berhad (Tan Sri Abdul Wahid bin Omar is the current board chairman of Cypark and former chairman of PNB)

DEAL IDEA

(15)

Deal Type	Asset	Rationale	Seller	Buyer
Special Purpose Vehicle (SPV) + Green Sukuk	SPV pools multiple small-to-medium scale Renewable Energy (RE) and Battery Energy Storage System (BESS) projects.	Developers (EPCC): - Overcomes current market fragmentation, allowing smaller players to collectively participate in larger utility-scale projects - Secure stronger negotiating power for off-take contracts and more favorable financing terms. - Synergizes complementary technical expertise across solar and BESS providers while minimizing overall financial exposure. Asset Owner/Sponsor: - Facilitates a strategic shift away from volatile, project-based EPCC contract revenues. - Equity injection into the SPV secures entitlement to long-term, highly stable recurring income streams. : Consolidating assets de-risks the balance sheet, significantly lowering the cost of capital for the Green Sukuk. Off-taker: - YTL's Johor Data Center Park provides a massive, built-in market to sell green energy directly to clients via CRESS. - Owning PowerSeraya allows YTL to export surplus renewable energy to Singapore through the ASEAN Power Grid (APG)	- Samaiden - Sunview - NuEnergy (BESS) - ERS Energy (BESS)	YTL Power - Strong AI infrastructure player - Own PowerSeraya in Singapore which could facilitate energy sales via APG

TARGETED COMPANIES

(16)

Company	Market Cap	Owned Capacity	Revenue Breakdown	Debt-to-Asset (2018-Current)	Ownership Structure	EV/EBITDA
Cypark Resources Berhad	RM 546.18 million	350-400 MW	65.24% Renewable Energy	0.4 - 0.74	21.47% Jakel Capital 8.37% Daud bin Ahmad	x27.95
Solarvest Holdings Berhad	RM 2.16 billion	200-350 MWp	85.54% EPCC	0.22 - 0.30	17.7% Atlantic Blue Holdings Sdn Bhd 6.48% Lim Chin Siu 6.37% Urusharta Jamaah Sdn Bhd	x32.20
Tenaga Nasional Bhd	RM 83.82 billion	4.6 GW	96.14% Electricity	0.43 - 0.45	23.44% EPF 18.05% Khazanah 13.12% PNB	x7.10
Samaiden Group Bhd	RM 497.19 million	200-300 MW	98.5% EPCC	0.01 - 0.24	29.31% Chow Pui Hee 18.48% Fong Yeng Foon 16.54% Chudenko Corp.	x10.32

Company	Market Cap	Owned Capacity	Revenue Breakdown (2025)	Debt-to-Asset (Historical 7 years)	Ownership Structure	EV/EBITDA
Gamuda Bhd	RM 22.33 billion	3 GW	77.6% Engineering & Construction	0.30 - 0.36	• 19.42% EPF • 8.10% PNB • 3.90% AIA Investment Management Private Limited	x20.02
YTL Power	RM 25.92 billion	500-600 MW	55.92% Power Generation	0.58 - 0.53	• 52.11% YTL Corp. • 10.92% EPF • 9.50% Yeoh Tiong Lay & Sons Family Holdings Limited	x8.58
NuEnergy	RM 104.84 million	16.33 MWac	75.84% Sales of Electricity	0.22 - 0.34	• 57.99% Agromate Holdings Sdn Bhd • 10.06% Liew Foo Heen • 3.52% HSBC Global Asset Management (UK) Ltd.	x31.11
Sunview Group Berhad	RM 208.89 million	100MW	93.6% EPCC	0.48 - 0.40	• 23.69% New Energy Capital Sdn Bhd • 4.14% Ong Hang Ping	x25.05

ZAVIER LEE

THANK
YOU